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REG NO : 2025-APC-82

BS APPLIED COMPUTING (B)

Topic: Food Delivery System

Purpose

The main purpose of a Food Delivery System is to make it easy for people to order food online. Customers can use the system to find restaurants, check their menus, select food, place an order, and get the food delivered to their home. The system also helps restaurants manage orders and allows delivery riders to deliver food to customers.

Users

The main users of a Food Delivery System are:

1. **Customers:** They search for restaurants, choose food, place orders, make payments, and track their orders.
2. **Restaurant Staff:** They receive customer orders, prepare the food, and update the order status.
3. **Delivery Riders:** They receive delivery requests, collect food from restaurants, and deliver it to customers.
4. **Administrators:** They manage the overall system, including users, restaurants, orders, payments, and complaints.

Stakeholders

The main stakeholders of the Food Delivery System are:

1. **Customers:** They want an easy and reliable way to order food.
2. **Restaurant Owners:** They use the system to receive online orders and want to grow their business.
3. **Delivery Riders:** They use the system to find delivery jobs and source of income.
4. **Food Delivery Company:** The company manages the platform and makes sure the service works properly.

5. **Payment Service Providers:** They help process online payments securely and fastly.
6. **System Administrators:** They are responsible for managing and maintaining the system.

Functional Requirements

1. **User Registration and Login:**
The system should allow customers, restaurant staff, and delivery riders to create an account and log in securely.
2. **Browse Restaurants and Food:**
Customers should be able to search for restaurants and view their menus, food items, prices, ratings, and availability.
3. **Place an Order:**
Customers should be able to select food items, add them to a cart, enter their delivery address, and place an order.
4. **Make Payment:**
The system should allow customers to pay for their orders using methods such as cash on delivery, credit cards, or mobile wallets.
5. **Track Order:**
Customers should be able to see the current status of their order, such as Order Placed, Preparing etc

Non-Functional Requirements

1. **Performance:**
The system should respond quickly when users search for restaurants, view menus, or place orders.
2. **Security:**
The system should keep customer information, passwords, and payment details safe from unauthorized access.
3. **Availability:**
The system should be available most of the time so customers can order food whenever they need it.

4. **Scalability:**

The system should be able to handle more customers, restaurants, and orders as the business grows.

5. **Usability:**

The application should be simple and easy to understand so that even a new user can place an order without difficulty.

Risks

1. **Payment Failure:**

Sometimes an online payment may fail because of internet problems, banking issues, or incorrect payment information. This can cause problems when placing an order.

2. **Delivery Delays:**

Food delivery can be delayed because of traffic, bad weather, wrong addresses, or not having enough delivery riders.

3. **Data Security Problems:**

There is a risk that hackers or unauthorized people could access customers' personal or payment information if the system is not properly secured.

Quality Attributes

1. **Reliability:**

The system should work correctly and process customer orders without losing or changing important information.

2. **Security:**

The system should protect the personal and payment information of customers, restaurants, and delivery riders.

3. **Usability:**

The system should have a simple and user-friendly design so customers can easily search for food, place orders, and track their deliveries.

4. **Good Rating:**

The system or the platform that are used by the delivery riders must be socially well reputable and have good ratings.

Where AI Could Be Used

AI can be used in different parts of a Food Delivery System to make the service more useful and efficient.

1. Food Recommendations:

AI can recommend restaurants and food items based on what a customer has ordered or searched for in the past.

2. Delivery Route Planning:

AI can study traffic and road conditions and suggest better routes to delivery riders.

3. Customer Support:

An AI chatbot can answer common customer questions about orders, payments, delivery times, and cancellations.

4. Predicting Food Demand:

AI can predict when there will be more orders. This can help restaurants prepare enough food and help the company arrange enough delivery riders.

What Could Go Wrong if AI Is Used Incorrectly?

1. Wrong Recommendations:

If the AI does not understand a customer's preferences correctly, it may recommend food that the customer does not like.

2. Wrong Delivery Routes:

If the AI has incorrect traffic or location information, it may suggest a bad route and cause the delivery to be late.

3. Incorrect Customer Support:

An AI chatbot might give a customer wrong information about their order, payment, refund, or cancellation.

4. Privacy Issues:

If customer information is not handled properly, AI systems could expose or misuse personal data.

Conclusion

A Food Delivery System makes ordering food easier by connecting customers, restaurants, and delivery riders through one platform. Customers can search for food, place orders, make payments, and track their deliveries. Restaurants can manage their orders, while riders can handle deliveries. AI can make the system even more useful through food recommendations, route planning, customer support, and demand prediction. However, AI

should be used carefully because incorrect information or poor use of customer data can create problems.